Usability testing

Lecture 7, IT Product Development

Innopolis University, 2026

Agenda

- Usability testing
- Running the test
- Post-test surveys
- Analyzing the results

Usability testing

Review

What is usability?

Why do you think you need to test it?

Usability testing

Adapted from Usability testing:

- **Goal:** Identify problems end-users can have before releasing the product, understand user behavior, get feedback to improve the product.
- **How:** Evaluate the product by testing it with representative users and observing their behavior.
- **What:**
  - Collect qualitative and quantitative data.
  - Determine the participant's satisfaction with the product.

Can be used to evaluate Interaction Capability (ISO/IEC 25010) of your product.

Usability testing: focus

Focus on:

- Measure
- Data
- Learn

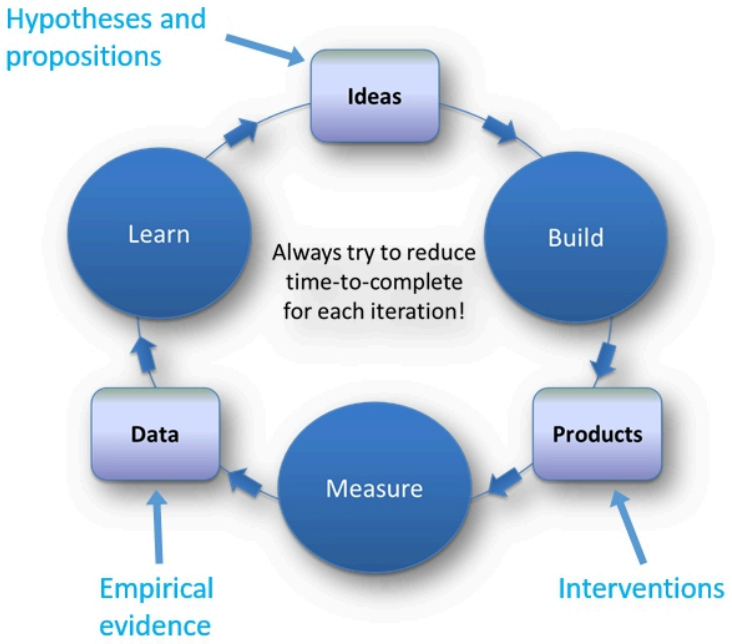


Figure 1: Ries' Lean startup, source

Usability test process outline

Adapted from Usability testing and How to run a usability test:

1. Select the testing goal and audience
2. Decide on the scope, budget, time
3. Establish evaluation criteria (measures)
4. Decide on tasks for the test
5. Instrument your system to record the relevant measures
6. Prepare scripts (instructions, tasks, questions, etc.)
7. Screen participants (filter out irrelevant participants)
8. Run pre-test survey
9. Run test
10. Run post-test survey
11. Organize data
12. Interpret and analyze
13. Learn

Usability testing vs User acceptance testing

Adapted from User Acceptance Testing vs Usability Testing:

- **Usability testing** - Used to uncover usability problems. Performed during the design and development.
- **User acceptance testing** - Used to uncover mismatches with requirements and expectations of stakeholders. Usually happens close to the final release of the product.

Key takeaways (Usability testing)

- Usability testing is used to uncover potential end-users’ problems before releasing the product to end-users.
- Usability testing requires finding real participants representing the end-users.
- Usability testing relies on qualitative and quantitative data, and you should plan in advance how you collect and evaluate it.

Running the usability test

Quiz

- What should you have scripted for a usability test?

Run the pre-test survey

Adapted from Common usability testing questions:

- **Goal:** Learn more about your participant before the test influences how they might answer.

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Examples of questions:

- What could you tell about your current role?
- What work-related mobile apps do you use?
- How familiar are you with …?

Run the test

Adapted from How user testing works:

- Give users tasks where they use the software without your assistance and think out loud.
- Ask (open-ended) questions, starting with more general ones and getting to more specific ones.
- For every task, ask for feedback after they complete that task.
- Observe what the user does and take notes.
- Record the user’s device screen to later be able to match user’s comments and visible interactions.

Run the post-test survey

Adapted from Follow-up usability test questions:

- Get feedback on the experience overall
- See if there’s something participants want to talk about that you didn’t ask

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Examples of questions:

- What was your overall impression of the experience?
- If you could change anything about it, what would you change?
- Is there anything we didn’t ask you about this experience that we should have?

Key takeaways (Run the usability test)

- Let participants complete specific tasks without your guidance, like real end-users would do.
- Ask questions before, during, and after the test.

Usability test surveys

Survey frameworks

Adapted from UX testing tools:

- Post-task (after a single task is completed):
  - Single Ease Question (SEQ)
  - After-Scenario Questionnaire (ASQ)
- Post-test (during the post-test survey)
  - System Usability Scale (SUS)
- Any stage:
  - Open-ended questions

Single Ease Question (SEQ) [post-task]

Overall, how difficult or easy was the task to complete?



After-Scenario Questionnaire (ASQ) [post-task]

1. Overall, I am satisfied with the ease of completing the task in this scenario.
2. Overall, I am satisfied with the amount of time it took to complete the task in this scenario.
3. Overall, I am satisfied with the support information (online help, messages, documentation) when completing the task.



System Usability Scale (SUS) [post-test]

Standard 10 questions in order:

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.

…

With the following response format:

Strongly Disagree 1	2	3	4	Strongly Agree 5
○	○	○	○	○

Source and details: https://measuringu.com/sus/

Open-ended questions

Goal: fill in missing context.

Ask more specific questions about new pieces of information that the user mentions.

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Possible approaches for designing questions:

- ORID
- The Mom Test

Key takeaways (Usability test surveys)

- Design questions to maximize learning and achieve the user test goals.
- Check established methods that enable relevant analysis.

Analyzing the results

Quiz

- Why need to analyze the data?

Organize: observation

Correct usability (Gibson affordances)	Exploratory actions (Norman affordances)
The user saw the search bar on the interface and used it	The user was using the Ctrl-F key to get to the “search” feature
…	…

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Adapted from Understanding affordances:

- **Gibson**: affordances exist whether or not they are perceived by the user – they exist independently, and therefore don’t change as the needs and goals of the user changes.
- **Norman**: affordances may exist in actuality, but they may also exist only in the perception of the user – based on previous experience or knowledge.
- Therefore, while one person may see an iPad as a useful device for checking email and watching YouTube, another might see it as a stylish cutting board.

Organize: match user

Match each user to the corresponding:

- pre-test survey
- specific task
- observations
- post-test survey

Recap: System Usability Scale (SUS)

Adapted from Measuring Usability with the System Usability Scale (SUS).

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…

With the following response format:

Strongly Disagree 1	2	3	4	Strongly Agree 5
○	○	○	○	○

Interpret and analyze: SUS example

Adapted from Measuring Usability with the System Usability Scale (SUS).

SUS = 2.5 · (20 + (sum of scores for odd questions) - (sum of scores for even questions))

Based on 500 studies, the “average” score is 68.

68 is actually 50th percentile - around 50% of studied products (accounting for ties at SUS 68) have SUS lower than or equal to 68.

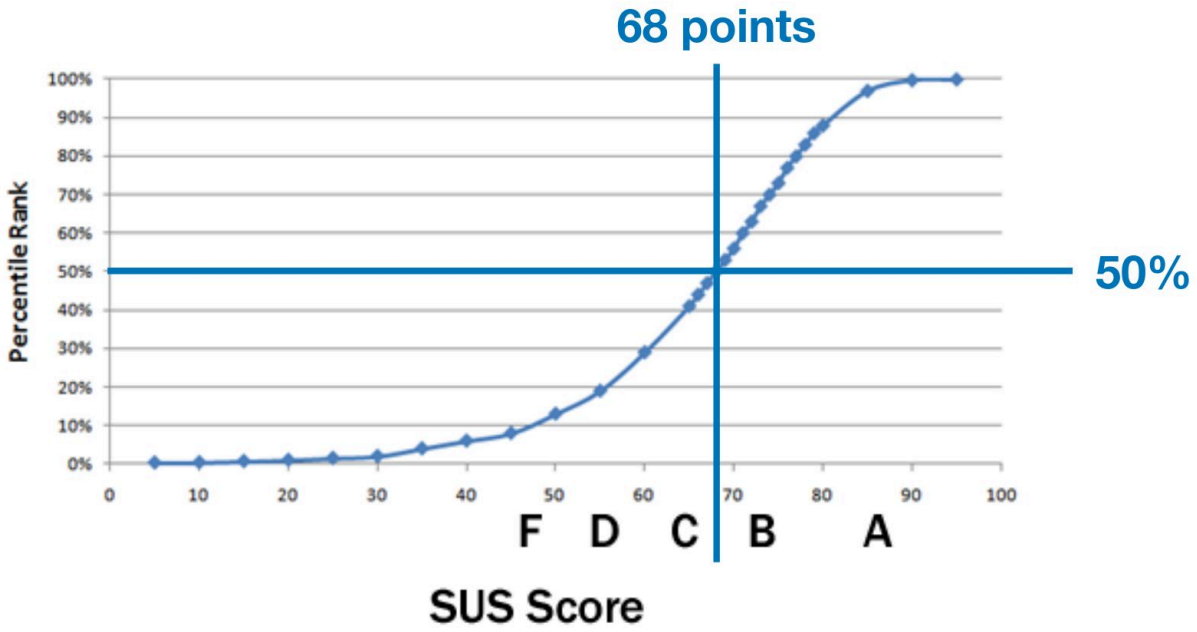
Below 68 is below average, above 80 is excellent.

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For a sample (several people), you can use the mean of SUS.

See Sample Sizes for Comparing SUS to a Benchmark

Interpret and analyze: SUS example



Learn

- Write down what you have learned after interpreting and analyzing the results.
- Decide what to change in your approach to designing user tests and to building the product.

Key takeaways (Analyze the results)

- Analyze the data to make data-driven decisions.
- Learn from what your observations and improve your procedures.

Quiz

What is usability?

Why do you think you need to test it?

What do we focus on in usability testing?

Describe the usability testing process.

Describe the surveys that we can use.

What is SUS?

How do we interpret SUS results?